

Problem

A series RLC circuit is described by

$$L \frac{d^2 i}{dt^2} + R \frac{di}{dt} + \frac{i}{C} = 10$$

Find the response when $L = 0.5 \text{ H}$, $R = 4 \text{ } \Omega$, and $C = 0.2 \text{ F}$. Let $i(0) = 1$, $di(0)/dt = 0$.

Step-by-step solution

Step 1 of 5

Write the differential equation that describes series RLC circuit.

$$L \frac{d^2 i}{dt^2} + R \frac{di}{dt} + \frac{i}{C} = 10$$

Divide equation with L on both sides.

$$\frac{L}{L} \frac{d^2 i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{i}{LC} = \frac{10}{L}$$

$$\frac{d^2 i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{i}{LC} = \frac{10}{L}$$

The solution to the differential equation is,

$$i(t) = i_t(t) + i_{ss}(t) \dots\dots (1)$$

Here,

$i_t(t)$ is the transient response

$i_{ss}(t)$ is the steady state response

Step 2 of 5

The circuit is a series RLC circuit.

Calculate the neper frequency.

$$\alpha = \frac{R}{2L}.$$

Substitute $4\ \Omega$ for R , 0.5 H for L .

$$\begin{aligned}\alpha &= \frac{R}{2L} \\ &= \frac{4}{2(0.5)} \\ &= 4\text{ s}^{-1}\end{aligned}$$

Calculate the natural frequency.

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Substitute 0.5 H for L , and 0.2 F for C .

$$\begin{aligned}\omega_0 &= \frac{1}{\sqrt{(0.5)(0.2)}} \\ &= 3.162\text{ rad/sec}\end{aligned}$$

Compare the values of α and ω_0 .

Since, $\alpha > \omega_0$ the system is over damped.

Step 3 of 5

The differential equation is,

$$\frac{d^2i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{i}{LC} = 0$$

Replace first derivative with s and second derivative by s^2 .

The characteristic equation is,

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0 \dots\dots (2)$$

Substitute $4\ \Omega$ for R , 0.5 H for L , and 0.2 F for C .

$$s^2 + \frac{4}{0.5}s + \frac{1}{(0.5)(0.2)} = 0$$

$$s^2 + 8s + 10 = 0$$

Calculate the roots of the characteristic equation.

$$\begin{aligned}s_{1,2} &= \frac{-8 \pm \sqrt{(8)^2 - (4)(1)(10)}}{(2)(1)} \\ &= \frac{-8 \pm 4.89}{2} \\ &= -1.55, -6.45\end{aligned}$$

Step 4 of 5

Calculate the transient response.

$$i_t(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

Substitute -1.55 for s_1 and -6.45 for s_2 .

$$i_t(t) = A_1 e^{-1.55t} + A_2 e^{-6.45t} \dots\dots (3)$$

Calculate the steady state response.

$$i_{ss}(t) = \frac{10}{L}$$

Substitute 0.5 H for L .

$$\begin{aligned} i_{ss}(t) &= \frac{10}{0.5} \\ &= 20 \end{aligned}$$

Recall equation (1).

$$i(t) = i_t(t) + i_{ss}(t)$$

Substitute $A_1 e^{-1.55t} + A_2 e^{-6.45t}$ for $i_t(t)$ and 20 for $i_{ss}(t)$.

$$i(t) = A_1 e^{-1.55t} + A_2 e^{-6.45t} + 20 \dots\dots (4)$$

Apply initial conditions. Substitute 0 for t .

$$i(0) = A_1 e^{-1.55(0)} + A_2 e^{-6.45(0)} + 20$$

$$i(0) = A_1 + A_2 + 20$$

Substitute 1 for $i(0)$.

$$1 = A_1 + A_2 + 20$$

$$A_1 + A_2 = -19 \dots\dots (5)$$

Step 5 of 5

Recall equation (4).

$$i(t) = A_1 e^{-1.55t} + A_2 e^{-6.45t} + 20$$

Differentiate equation with respect to t .

$$\frac{di(t)}{dt} = -1.55 A_1 e^{-1.55t} - 6.45 A_2 e^{-6.45t}$$

Apply initial conditions. Substitute 0 for t .

$$\frac{di(0)}{dt} = -1.55 A_1 e^{-1.55(0)} - 6.45 A_2 e^{-6.45(0)}$$

$$\frac{di(0)}{dt} = -1.55 A_1 - 6.45 A_2$$

Substitute 0 for $\frac{di(0)}{dt}$.

$$0 = -1.55 A_1 - 6.45 A_2 \dots\dots (6)$$

Solve equations (5) and (6).

$$A_1 = -25$$

$$A_2 = 6$$

Recall equation (4).

$$i(t) = A_1 e^{-1.55t} + A_2 e^{-6.45t} + 20$$

Substitute -25 for A_1 , 6 for A_2 .

$$\begin{aligned} i(t) &= -25e^{-1.55t} + 6e^{-6.45t} + 20 \text{ A} \\ &= 20 - 25e^{-1.55t} + 6e^{-6.45t} \text{ A} \end{aligned}$$

Therefore, the response, $i(t)$ of series RLC circuit is $\boxed{20 - 25e^{-1.55t} + 6e^{-6.45t} \text{ A}}$.